WORKED SOLUTION

STEP I 1992, Question 2

Pure • Magic Squares and Counting

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Read the question before you go on. It is in the STEP Support Programme's question database:

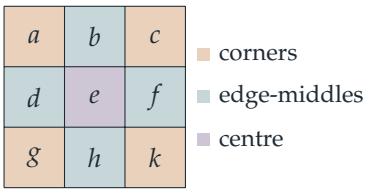
step.maths.org/questions/92-s1-q2

The key idea

Everything comes from adding up lines cleverly. Adding all three rows uses each number exactly once, which gives n. Adding the four lines through the centre uses every number once and the centre four times, which pins down e. After that the question becomes a small, careful case analysis: once e is known, each line through the centre pairs up two numbers with a fixed sum, and that leaves very little room for the large numbers. Symmetry of the square (rotating it) then does the counting for us.

Solution

It helps to name the three kinds of cell: the four *corners* a, c, g, k, the four *edge-middles* b, d, f, h, and the *centre* e.



Part (i)

The three rows together contain each of the nine entries exactly once. So, adding the three row sums,

$$3n = (a+b+c)+(d+e+f)+(g+h+k) = 1+2+\cdots+9 = 45,$$

and hence $n = 15$.

Part (ii)

Four lines pass through the centre: the middle row, the middle column and the two diagonals. Adding them,

$$(d+e+f)+(b+e+h)+(a+e+k)+(c+e+g)=4n=60.$$

On the left, every entry other than e appears exactly once and e appears four times, so the left-hand side is $45+3e$. Therefore $45+3e=60$, giving $e=5$.

A useful consequence: since each line through the centre sums to 15 and $e=5$, the two cells *opposite* each other through the centre always sum to 10:

$$a+k=c+g=b+h=d+f=10.$$

Part (iii)

The entry 9 is not the centre, because $e=5$. So we must show that 9 cannot be in a corner.

Suppose, for contradiction, that 9 is in a corner. The argument is the same for every corner, so take $a=9$. Then the opposite corner is $k=10-9=1$. The top row and the left column both pass through a, so

$$b+c=15-9=6 \qquad \text{and} \qquad d+g=15-9=6.$$

The four numbers b,c,d,g are distinct and are drawn from what is left once $1,5,9$ are used, namely $\{2,3,4,6,7,8\}$. The only way to write 6 as a sum of two *different* numbers from this set is $2+4$ (since $1+5$ uses numbers already placed and $3+3$ repeats). So $\{b,c\}=\{2,4\}$ and $\{d,g\}=\{2,4\}$, which needs the numbers 2 and 4 twice each. That contradicts distinctness.

? **Ask yourself** Why does it suffice to deal with the corner a alone?

For any corner, the same three facts hold: the opposite corner must be 1, and the two lines along the edges of the square through that corner each need their other two entries to sum to 6. So the same contradiction appears. (Equivalently, rotating a magic square through a quarter turn gives another magic square, and rotations carry a to each of the other corners.)

So 9 is neither the centre nor a corner, and it must be one of b,d,f,h.

Part (iv)

Let $b=9$. Then $h=10-9=1$, and the top row gives $a+c=6$. As in (iii), the unused numbers are $\{2,3,4,6,7,8\}$, and the only pair of distinct numbers from them with sum 6 is $\{2,4\}$. So either $a=2,\ c=4$ or $a=4,\ c=2$.

Once a and c are fixed, everything else is forced by opposite pairs and the side columns:

- $a=2,\ c=4$: then $k=10-a=8$, $g=10-c=6$, $d=15-a-g=7$ and $f=15-c-k=3$.

- $a = 4$, $c = 2$: then $k = 6$, $g = 8$, $d = 15 - 4 - 8 = 3$ and $f = 15 - 2 - 6 = 7$.

Forcing values only shows these are the *only* candidates; we must still check that they work. In each case the entries are $1, \ldots, 9$ once each, and every row, column and diagonal sums to 15:

2	9	4
7	5	3
6	1	8

4	9	2
3	5	7
8	1	6

(For example, in the first: rows $15, 15, 15$; columns $2 + 7 + 6$, $9 + 5 + 1$, $4 + 3 + 8$; diagonals $2 + 5 + 8$, $4 + 5 + 6$.) So there are exactly two magic squares with $b = 9$, and the second is the mirror image of the first in the middle column.

Part (v)

Turning a magic square through a quarter turn clockwise gives another magic square: its rows are the old columns, its columns are the old rows (read in reverse), and the two diagonals swap. The quarter turn is reversible (turn back anticlockwise), so it gives a one-to-one correspondence between magic squares.

A quarter turn clockwise moves the cell b to the cell f. So the magic squares with $f = 9$ are exactly the quarter-turns of the magic squares with $b = 9$, and there are exactly 2 of them. Turning again, there are exactly 2 with $h = 9$ and exactly 2 with $d = 9$.

By (iii), every magic square has 9 in exactly one of the positions b, f, h, d, so these four groups are disjoint and cover everything. Hence there are

$$4 \times 2 = 8$$

different 3×3 magic squares. They are the two squares in (iv) together with their rotations through a quarter, half and three-quarter turn; equivalently, they are the eight rotations and reflections of a single square.

Checks

A short computer search over all $9! = 362\,880$ arrangements of $1, \ldots, 9$ finds exactly 8 magic squares. All of them have line sum 15 and centre 5, all have 9 in an edge-middle, and the two with $b = 9$ are the squares found in (iv). By hand, you can check that in every square 1 sits opposite 9 and the even numbers $2, 4, 6, 8$ fill the corners.

Common slips

- In (iv), forgetting to check that the forced squares really are magic. The deduction only shows that nothing else can work.
- In (v), answering "2" or "4" without explaining why rotation produces new, genuinely different squares, or without using (iii) to see that the four cases cover everything without overlap.

Beyond STEP

Part (v) is a first taste of *group actions*: the eight symmetries of the square (four rotations and four reflections) form a group, and the magic squares make up a single orbit under it. Counting objects up to symmetry like this is developed in Burnside's lemma, which you meet in a first course in group theory. Allenby's *Rings, Fields and Groups* is a friendly place to start.

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